



Linux versioning scheme and development process

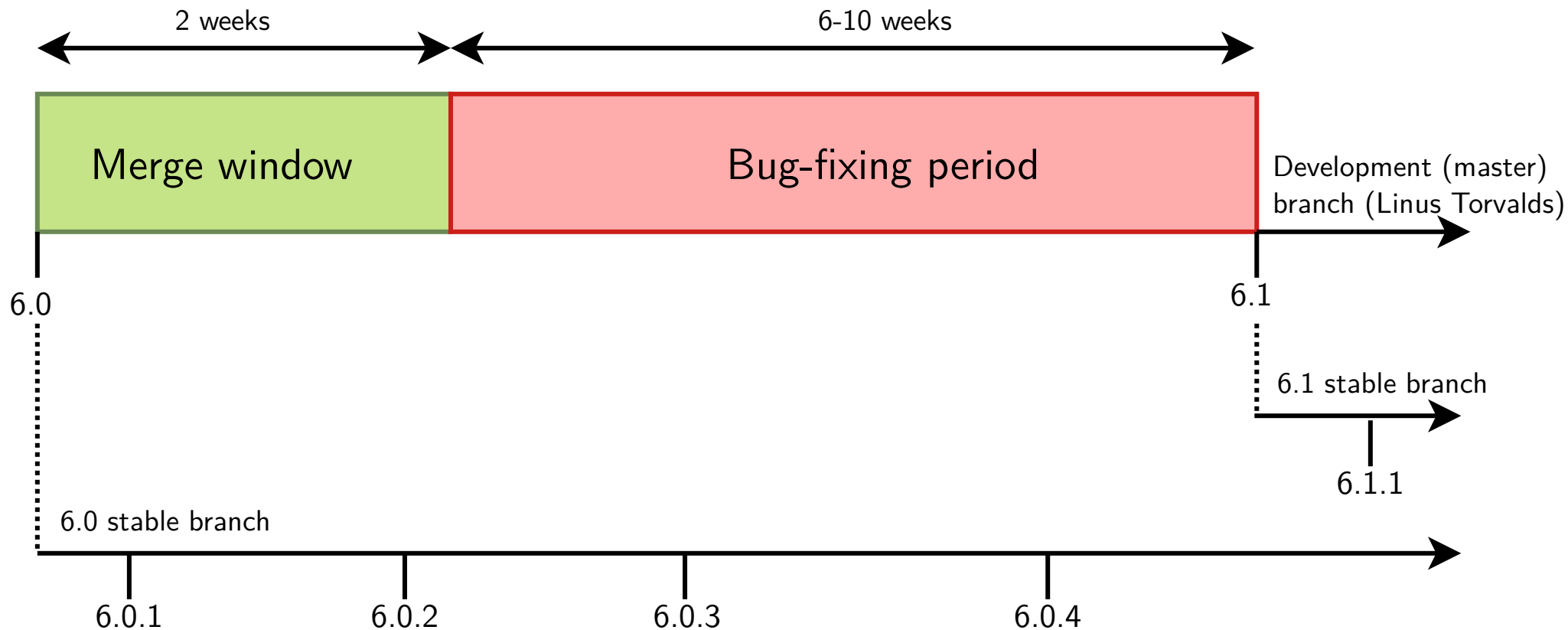


Linux versioning scheme

- ▶ Until 2003, there was a new “stabilized” release branch of Linux every 2 or 3 years (2.0, 2.2, 2.4). Development branches took 2-3 years to be merged (too slow!).
- ▶ Since 2003, there is a new official release of Linux about every 10 weeks:
 - Versions 2.6 (Dec. 2003) to 2.6.39 (May 2011)
 - Versions 3.0 (Jul. 2011) to 3.19 (Feb. 2015)
 - Versions 4.0 (Apr. 2015) to 4.20 (Dec. 2018)
 - Versions 5.0 (Mar. 2019) to 5.19 (July 2022)
 - Versions 6.0 (Oct. 2022) to 6.19 (Feb. 2026)
 - Version 7.0 will be released in March/April 2026.
- ▶ Features are added to the kernel in a progressive way. Since 2003, kernel developers have managed to do so without having to introduce a massively incompatible development branch.
- ▶ For each release, there are bugfix and security updates called stable releases: 7.0.1, 7.0.2, etc.



Linux development model Using merge and bug fixing windows





Need for long term support (1)

- ▶ Issue: bug and security fixes only released for most recent kernel versions.
- ▶ Solution: the last release of each year is made an LTS (*Long Term Support*) release, and is supposed to be supported (and receive bug and security fixes) for at least 2 years.

Version	Maintainer	Released	Projected EOL
6.18	Greg Kroah-Hartman & Sasha Levin	2025-11-30	Dec, 2027
6.12	Greg Kroah-Hartman & Sasha Levin	2024-11-17	Dec, 2026
6.6	Greg Kroah-Hartman & Sasha Levin	2023-10-29	Dec, 2026
6.1	Greg Kroah-Hartman & Sasha Levin	2022-12-11	Dec, 2027
5.15	Greg Kroah-Hartman & Sasha Levin	2021-10-31	Dec, 2026
5.10	Greg Kroah-Hartman & Sasha Levin	2020-12-13	Dec, 2026

Captured on <https://kernel.org> in Feb. 2026, following the [Releases](#) link.

- ▶ Example at Google: starting from *Android O (2017)*, all new Android devices have to run such an LTS kernel.



Need for long term support (2)

- ▶ You could also get long term support from a commercial embedded Linux provider.
 - Wind River Linux can be supported for up to 15 years.
 - Ubuntu Core can be supported for up to 10 years.
- ▶ *“If you are not using a supported distribution kernel, or a stable / longterm kernel, you have an insecure kernel”* - Greg KH, 2019 Some vulnerabilities are fixed in stable without ever getting a CVE.
- ▶ The *Civil Infrastructure Platform* project is an industry / Linux Foundation effort to support much longer (at least 10 years) selected LTS versions (currently 4.4, 4.19, 5.10 and 6.1) on selected architectures. See <https://wiki.linuxfoundation.org/civilinfrastructureplatform/start>.